

Recurrence Formula for an Integral

Topic.

Integration by parts, recurrence formulas for integrals.

Statement.

Using integration by parts, obtain a recurrence formula for the integral

$$A_n = \int \frac{x^n}{\sqrt{x^2 + a}} dx, \quad n > 2.$$

Solution.

Let

$$A_n = \int \frac{x^n}{\sqrt{x^2 + a}} dx, \quad n > 2.$$

Use integration by parts.

Take

$$u = x^{n-1}, \quad du = (n-1)x^{n-2} dx,$$

and

$$dv = \frac{x}{\sqrt{x^2 + a}} dx.$$

Then

$$v = \sqrt{x^2 + a},$$

because

$$\left(\sqrt{x^2 + a}\right)' = \frac{x}{\sqrt{x^2 + a}}.$$

Therefore,

$$A_n = uv - \int v du.$$

Hence,

$$A_n = x^{n-1}\sqrt{x^2 + a} - (n-1) \int x^{n-2}\sqrt{x^2 + a} dx.$$

Now transform the integral

$$\int x^{n-2} \sqrt{x^2 + a} \, dx.$$

We have

$$x^{n-2} \sqrt{x^2 + a} = x^{n-2} \cdot \frac{x^2 + a}{\sqrt{x^2 + a}}.$$

Therefore,

$$x^{n-2} \sqrt{x^2 + a} = \frac{x^n}{\sqrt{x^2 + a}} + a \frac{x^{n-2}}{\sqrt{x^2 + a}}.$$

So

$$\int x^{n-2} \sqrt{x^2 + a} \, dx = A_n + aA_{n-2}.$$

Substitute this into the formula for A_n :

$$A_n = x^{n-1} \sqrt{x^2 + a} - (n-1)(A_n + aA_{n-2}).$$

Now express A_n .

$$A_n + (n-1)A_n = x^{n-1} \sqrt{x^2 + a} - (n-1)aA_{n-2}.$$

Thus,

$$nA_n = x^{n-1} \sqrt{x^2 + a} - (n-1)aA_{n-2}.$$

Finally,

$$A_n = \frac{x^{n-1} \sqrt{x^2 + a}}{n} - \frac{(n-1)a}{n} A_{n-2}, \quad n > 2.$$

The base cases are

$$A_0 = \int \frac{dx}{\sqrt{x^2 + a}},$$

and

$$A_1 = \int \frac{x}{\sqrt{x^2 + a}} \, dx = \sqrt{x^2 + a} + C.$$